

BY GARY TAUBES

The experiment
ist Hideki Shirak
one of his students at the
Tokyo Institute of Technol
ogy to perform was routine: turn a
tiny amount of common acetylene
welding gas into an organic poly
mer, a plastic consisting of a long
chain of carbon-based molecules.

But the student, a Korean with a
limited Japanese vocabulary, mis
understood his professor and used
far too much of a crucial chemical
ingredient. Instead of obtaining a black powder, the
usual result in the experiment, he got a silvery film
that looked like nothing so much as a metal.

The strange material born of that botched experi
ment in 1970 might have remained a simple labora
tory curiosity had not Alan MacDiarmid, a Universi
ty of Pennsylvania chemist, traveled to Japan five
years later. When Shirakawa showed him the met
al-like plastic, he was intrigued. "I had never seen
anything like it before," MacDiarmid says. He was
so curious, in fact, that he invited Shirakawa to join
him and a colleague, physicist Alan Heeger, for a
year in their Philadelphia laboratory to investigate
the properties of the puzzling substance.

This collaboration soon led to another surprising
discovery. When a researcher intentionally contami
nated, or "doped," the new plastic with a small
amount of iodine—a standard chemical practice—the
properties of the polymer were altered radically.
This doping had increased the plastic's ability to con
duct electricity as much as a trillion-fold. For the first
time, scientists had created a plastic that was as good
an electrical conductor as metal—and, in the process,
opened a pathway to a world of electrical and elec



CHEMISTRY

Polyacetylene's shimmer shows off
its amazing metal-like properties

tronic components made of plastic.

The exotic material defied all the
conventional wisdom about plas
tics. Ever since John Hyatt, an
American inventor, created the
first synthetic polymer, celluloid, in
1868, plastics have been known for
their highly useful characteristics.
They are lightweight, pliable, easy
to mold, and excellent electrical in
sulators. For these reasons, plastics
have long been used to coat electri

cal wires and to make exterior parts of such com
mon electrical equipment as telephones, power
tools, and computer terminals.

MacDiarmid and other scientists quickly recog
nized the technological promise of plastics that were
also electrical conductors. Such materials, they
thought, might readily lend themselves to a host of
novel products. One application, in particular, has
since drawn the attention of a number of manufac
turers: a super-lightweight battery that might, at
long last, make electric cars both practical and eco
nomical. Until now, development of these quiet,
non-polluting vehicles has been hindered by their re
liance on conventional batteries, usually made of
lead, which quickly expend their energy and weigh
as much as all the other parts of the car combined.

Lured by the potential of a plastic battery and
other plastic devices, chemists everywhere have
been busily looking into the new material. The num
ber of scientific papers on conducting plastics has
proliferated from the original offering by MacDiar
mid and Heeger in 1977 to a flood of more than 200
last year. In recent years, chemists at IBM, Allied
Corporation, Chevron Chemical, Xerox, and several

ELECTRIFYING PLASTICS

*Plastics are not supposed to conduct
electricity, but the surprising discovery that
some can may open the door to battery-
powered cars and other futuristic technology*



In Alan MacDiarmid's laboratory, an experimental battery made of conducting plastic spins a propeller. Such plastics could power electric cars of the future

laboratories overseas have concocted their own variations of the polymers. Many researchers have used these substances to build experimental electronic gadgetry, including diodes and other solid state devices. Such components, usually made either of silicon or other semiconductors, are at the heart of microchips and the computer revolution. American industry alone is investing some \$5 million to \$10 million a year in research and development involving the remarkable plastics.

Unlike plastics, metals conduct easily because they have a surplus of free electrons that can readily be passed from one atom to the next. By contrast, the atoms of an insulator hold their electrons very tight. In the language of chemistry, the electrons are tightly bound to the nuclei of the atoms. Only a powerful jolt of external energy will free electrons so they can move on to neighboring atoms. Simon Porter, a chemist at MacDiarmid's laboratory, compares conductivity to a parking lot and electrons to cars: "If the parking lot is full, you have nowhere to go; there is no movement of cars. On the other hand, if a car leaves the lot by one exit, you can go in from the other side and park. There's movement of cars through the lot or, if you will, electrons. That's conduction."

But plastics have tightly bound electrons. How does polyacetylene, the plastic derived from acetylene gas, manage to be-

come a conductor? One possible explanation lies in its unique structure, the simplest of any plastic. It is produced with the aid of a special catalyst that encourages the free-floating molecules in the gas to link up in chains, or polymers. Excessive amounts of this catalyst, as Shirakawa discovered, result in longer polymers and impart the metallic look to the plastic.

Each of the identical, linked units in the polymer consists of a carbon atom and hydrogen atom. Along the chain, the adjacent carbon atoms are bound to each other by shared electrons. These bonds alternate: a single bond, to which each carbon atom contributes one electron, is followed by a double bond, to which each contributes two elec-



As physicist Alan Heeger looks on, researcher Joan Brennan, wearing rubber gloves, manipulates a piece of conducting plastic that is protected in an atmosphere of pure argon

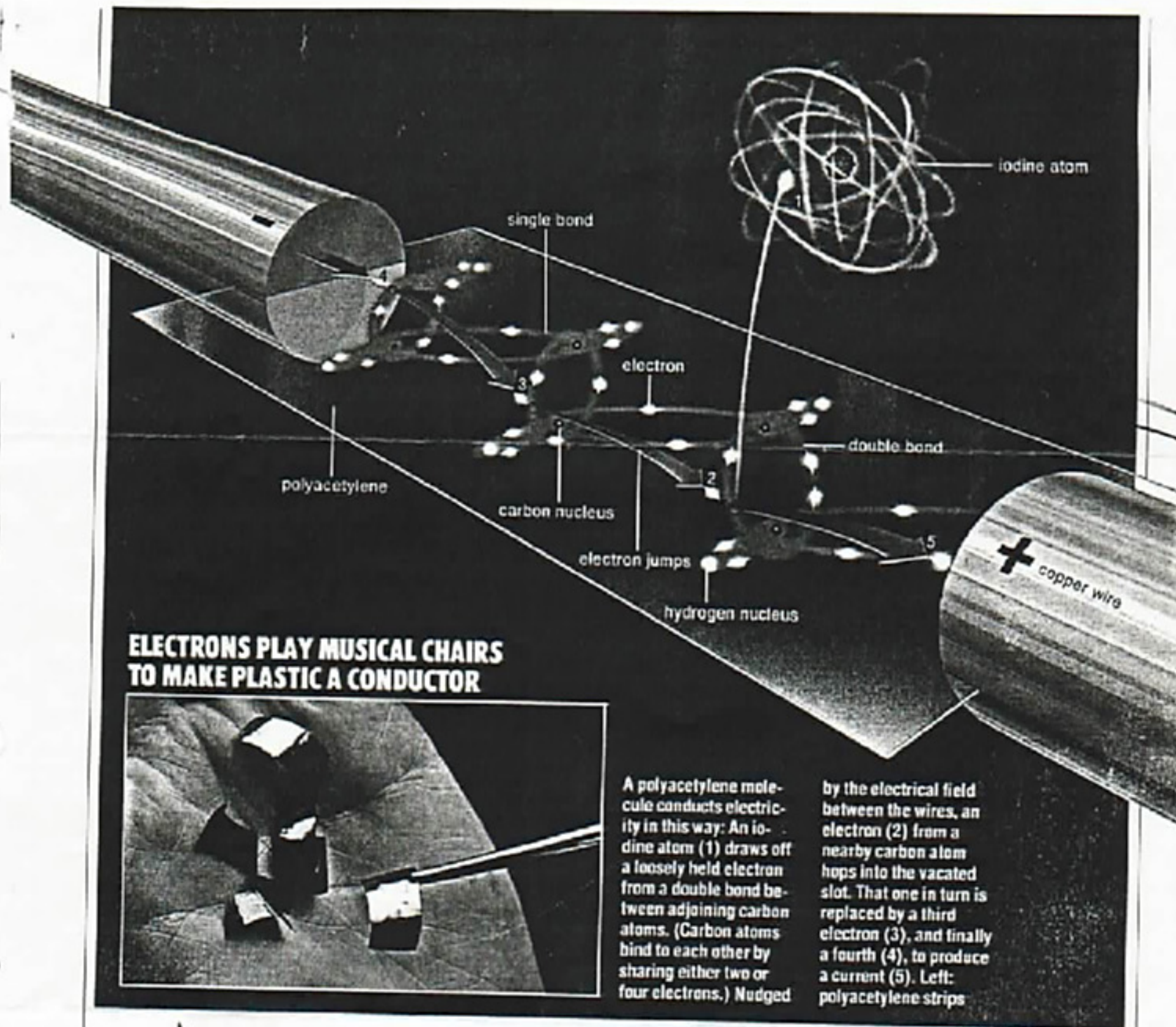
trons. In the double bonds, the electrons are loosely held by the nuclei of the atoms, and can be pried loose with the right chemical crowbar.

For polyacetylene, the crowbar was the doping substance, iodine, which can steal extra electrons from other atoms. The result of this theft is an occasional carbon atom in the chain that lacks a full complement of electrons. The missing electrons are in effect the empty parking spaces in Porter's analogy. When an external electrical field is applied to the plastic, nearby electrons promptly jump into the vacant slots. These in turn leave new vacancies that are filled by still other electrons. Suddenly, what was originally an insulator has become a conductor, allowing the passage of an electrical current.

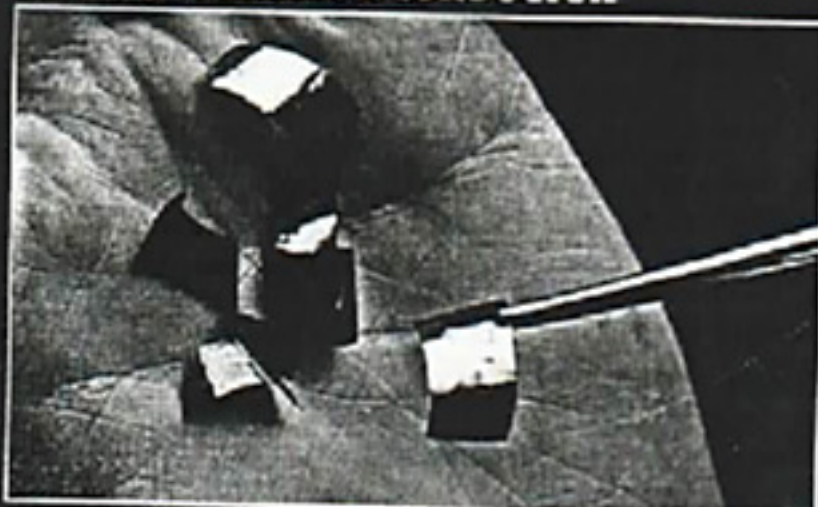
Taking advantage of this conductivity, industrial researchers have already produced scaled-down experimental plastic batteries that weigh a third less than comparable conventional batteries. Equally important, these plastic cells can be charged and discharged much faster than conventional lead-acid batteries. In an electric car, such rapid delivery of power would vastly improve acceleration and hill-climbing ability, both sorely deficient in today's electric vehicles. In addition, scientists report that their prototype polyacetylene batteries are highly durable, capable of being completely discharged and recharged more than a thousand times before wearing out. A typical lead-acid battery might not recover from even a single complete discharge.

What makes the plastic batteries so apparently superior? Like water tumbling down a mountainside, electrical currents flow through a battery because there is a difference in potential, or voltage, between its electrodes; this is comparable to different elevations at either end of the mountain stream. In ordinary batteries the electrical potential is created by dissimilar materials—say, lead and lead oxide—separated by a solution, such as sulphuric acid, called an electrolyte. When the battery is discharged, the electrodes gradually dissolve into positive ions (atoms that have lost electrons and thus have a positive charge) that move back and forth through the electrolyte. When the battery is charged, the ions return to the electrodes, pick up their missing electrons, and become full-fledged lead atoms again, replenishing the depleted electrodes. But after repeated cycles, the electrodes become coated with a crust of lead sulphate, and the battery fails.

By contrast, plastic batteries consist of two strips of polymer immersed in the



ELECTRONS PLAY MUSICAL CHAIRS TO MAKE PLASTIC A CONDUCTOR



A polyacetylene molecule conducts electricity in this way: An iodine atom (1) draws off a loosely held electron from a double bond between adjoining carbon atoms. (Carbon atoms bind to each other by sharing either two or four electrons.) Nudged

by the electrical field between the wires, an electron (2) from a nearby carbon atom hops into the vacated slot. That one in turn is replaced by a third electron (3), and finally a fourth (4), to produce a current (5). Left: polyacetylene strips

ILLUSTRATION BY GEORGE V. KELVIN

electrolyte. Unlike the process in conventional batteries, the electrolyte—rather than the electrodes themselves—contributes the ions that shuttle between the two electrodes. As the ions cling to the electrodes, they dope the polymer and change its potential. Because the ions are drawn from the electrolyte, not from the polymer, the electrodes have virtually no wear and tear and remain intact through countless charging and discharging cycles. Says chemist Fred Wudl, of the University of California at Santa Barbara, "Theoretically, you should be able to go on using the battery forever."

So far these plastic marvels have been confined to the laboratory. But Allied Corporation, which bought licensing rights from the University of Pennsylvania in

1981 to manufacture plastic batteries, hopes to bring a commercial version to the market within the next several years.

Cars are not the only potential market for the battery makers. They see a full range of products, from rechargeable batteries for appliances like video cassette recorders to huge storage batteries that could be used by public utilities to store electricity from wind or hydroelectric generating facilities during periods of low power demand. Industrial chemists are also thinking of using the new plastics to fabricate inexpensive solar panels that can convert sunlight directly into electricity. Up to now, such panels have usually been made of much costlier materials. MacDiarmid envisions immense sheets of plastic solar

cells and batteries rolling off production lines and being cut to the desired size and voltage by nothing more than mechanized cookie cutters.

Indeed, as chemists learn to tailor-make the plastics with specific chemical and physical properties, they should be able to produce a host of new electrically conducting polymers that could be used in place of costlier metals in everything from motors and wires to magnets and generators. "Our dream," says Heeger, "is to make materials that will create a whole new technology. After all, synthetic fibers like nylon didn't exist fifty years ago, and now they have replaced natural fibers in many applications. These new man-made metals may have the same promising future."